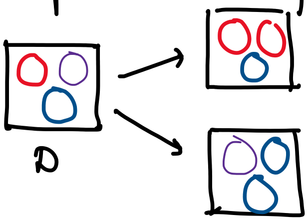
# Lecture 22

Saturday, May 2, 2026 9:21 AM

Recap: Bagging (Bootstrap Aggregating)

Bootstrapping: Simulating having multiple datasets by sampling with replacement from the given dataset.



$D$ $\tilde{D}_1$ $\tilde{D}_2$

Bagging

1. For $b = 1, 2, \dots B$
   - → a) Create dataset $\tilde{D}_b$ by bootstrapping
   - → b) Train model on $\tilde{D}_b$ (say a decision tree)
2. Prediction
   - → Regression: $\hat{y}$ = average of $B$ models
   - → Classification: $\hat{y}$ = majority vote of $B$ models

Limitations of Bagging with trees as models:
If one feature has very low Gini Impurity index when used as a split, all $B$ trees will use it for the root node.
We want tree to have diversity.

Fix: Random Forests: Bagging + Decision trees + feature subsampling at each node

| | Decision trees | Random Forest |
|---|---|---|
| Pros | Easy to interpret | Harder to intepret |
| Cons | poor performance | often much better |

Boosting (reference material will be provided)

Supervised learning
- $(x^{(i)}, y^{(i)})$ pairs
  $x^{(i)} \in \mathbb{R}^d$ $y^{(i)} \in \mathbb{R}$

Unsupervised learning.
- $x^{(i)}$ only

- learn $f: x \xrightarrow{J} y$

- Discover structure in $\{x^{(i)}\}_{i=1}^{N}$

Unsupervised learning
- no ground truth
- subjective

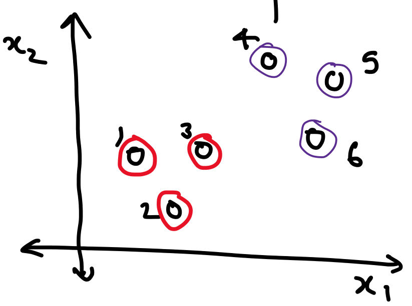
Clustering

- Defn: Clusters — set a data points with "similar" features. Similarity is measured by a distance metric — Euclidean distance

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \quad \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \qquad \sqrt{(x_1-y_1)^2 + (x_2-y_2)^2}$$

- Clustering : $C : \{1, 2, \dots N\} \rightarrow \{1, 2, \dots k\}$
  a function that assigns each data point $i$ to a ~~#~~ cluster

$d = 2$




$k = 2$

| data | cluster index |
|---|---|
| 1 | 1 |
| 2 | 1 |
| 3 | 1 |
| 4 | 2 |
| 5 | 2 |
| 6 | 2 |

Application : $N = 500{,}000$ customers
$d = 20 \leftarrow$
$K = 10$

- Clusters will group similar customers in a group
- Targetted marketing for each group

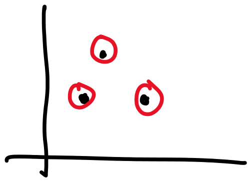
Formal Problem Setup.

- Dataset $\{x_1, x_2 \dots x_N\}$ $x_i \in \mathbb{R}^d$
- Integer $k$ : desired number of clusters

Output : Partition of the N points into k non-overlapping groups

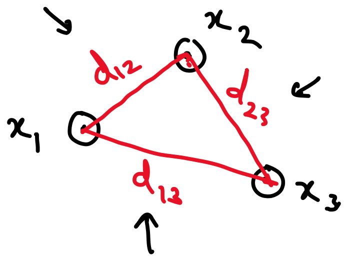
Center of each sub-group: average value of the feature vectors of the data-points in that group

Objective = ??
Minimizing within cluster spread



$C_l$ – cluster
$m_l$ – points in the cluster

$$\underset{\text{spread}}{W(C_l)} = \underbrace{\frac{1}{m_l}}_{\text{\# of points in } C_l} \sum_{i,j \in C_l} \underbrace{\|x_i - x_j\|^2}_{\text{square of euclidean distance between } x_i \,\&\, x_j}$$




$$W(C_l) = \frac{1}{3}\left(d_{12}^2 + d_{23}^2 + d_{31}^2\right)$$

$\underbrace{\text{Total spread}}_{\text{Objective function}}$ : $\underset{\text{clustering function}}{W(C)} = \sum_{l=1}^{k} W(C_l)$

To find clusters such that $W(C)$ is minimized

How is k determined? (next class)

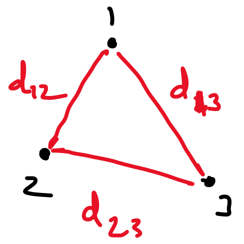
Center of data (??)

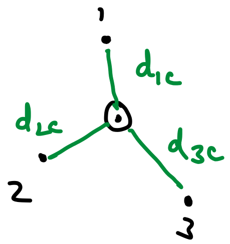
Theorem:
- $C_l$ be a ~~class~~ cluster of $m_l$ points.
- $\mu_l = \frac{1}{m_l} \sum_{i \in C_l} x_i$ ← center, centroid

$$\sum_{i,j \in C_l} \|x_i - x_j\|^2 = 2 m_l \sum_{i \in C_l} \|x_i - \mu_l\|^2$$

sum of pairwise squared ~~distance~~ distances

sq. distance of $x_i$ from centroid




$$d_{12}^2 + d_{13}^2 + d_{23}^3 = 6\left(d_{1c}^2 + d_{2c}^2 + d_{3c}^2\right)$$

Objective.

$$\underbrace{\min W(C)} = \boxed{\min \; SSE(C, \mu)}$$

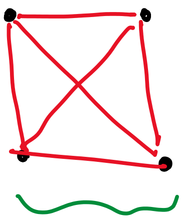
$$= \min \sum_{i=1}^{N} \| x_i - \mu_{c(i)} \|^2$$

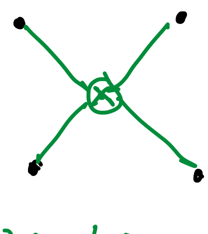
$x_i$: point

$\mu_{c(i)}$: centroid of the ~~cust~~ cluster containing $\mu_{c(i)}$

$$\sum_{l=1}^{K} W(C_l)$$

$$\mu_{c_l} = \frac{1}{m_l} \sum_{i \in C_l} x_i$$





$O(N^2)$

$\underbrace{O(N)}_{\text{centroid computation}} + \underbrace{O(N)}_{\text{distance computation}}$